Test Plan

By

Zainab Hassan Al-Badi	**2220002372**
Hebah Ahmed Altheyab	**2220001030**
Doja Abdulmohsen Alnemer	**2220002855**
Zinab Mukhtar Alrashed	**2220003194**
Donia Abdullah Alfarhan	**2220004173**

Supervised by
Mrs. Albandary Mohammad Alamer

Committee Member Names
Dr. May Issa Aldossary
Mrs. Hina Gull

1 March 2026

TABLE OF CONTENTS

1. INTRODUCTION 1

1.1 OBJECTIVES 1

1.2 TESTING STRATEGY 1
1.3 SCOPE 2
1.4 REFERENCE MATERIAL 3

1.5 DEFINITIONS AND ACRONYMS 3

2. TEST ITEMS 7

2.1 PROGRAM MODULES 7

2.2 JOB CONTROL PROCEDURES 7
2.3 USER PROCEDURES 7
2.4 OPERATOR PROCEDURES 7

3. FEATURES TO BE TESTED 8

4. FEATURES NOT TO BE TESTED 12

5. APPROACH 12

5.1 COMPONENT TESTING 13

5.2 INTEGRATION TESTING 18
5.3 CONVERSION TESTING 23
5.4 JOB STREAM TESTING 23
5.5 INTERFACE TESTING 23
5.6 SECURITY TESTING 23
5.7 RECOVERY TESTING 23
5.8 PERFORMANCE TESTING 23
5.9 REGRESSION TESTING 24
5.10 ACCEPTANCE TESTING 24
5.11 BETA TESTING 24

6. PASS / FAIL CRITERIA 24

6.1 SUSPENSION CRITERIA 25
6.2 RESUMPTION CRITERIA 25
6.3 APPROVAL CRITERIA 25

7. TESTING PROCESS 26

7.1 TEST DELIVERABLES 26
7.2 TESTING TASKS 26
7.3 RESPONSIBILITIES 26
7.4 RESOURCES 27
7.5 SCHEDULE 27

8. ENVIRONMENTAL REQUIREMENTS 28

8.1 HARDWARE 28
8.2 SOFTWARE 29
8.3 SECURITY 29

8.4 TOOLS 29
8.5 PUBLICATIONS 30
8.6 RISKS AND ASSUMPTIONS 31

9. CHANGE MANAGEMENT PROCEDURES 32

9.1 CHANGE INITIATION: 32
9.2 CHANGE REVIEW: 32
9.3 CHANGE AUTHORIZATION PROCESS: 32

10. PLAN APPROVALS 33

TABLE OF TABLES

Table 1: Definitions and Acronyms 3
Table 2: Features To Be Tested 8
Table 3 : Features Not to Be Tested 12
Table 4: Ai Diagnosis Module Successful Test Case 13
Table 5: Ai Diagnosis Module Unsuccessful Test Case 13
Table 6: Consultation Module: Booking Successful Test Case 14
Table 7: Consultation Module: Booking Unsuccessful Test Case 14
Table 8: Consultation Module: Management Successful Test Case 15
Table 9: Consultation Module: Chat System Successful Test Case 16
Table 10: Consultation Module: Chat System Unsuccessful Test Case 16
Table 11: Knowledge Hub (Mini Wiki) Module Successful Test Case 17
Table 12: History Module Successful Test Case 17
Table 13: History Module Unsuccessful Test Case 18
Table 14: Role selection Successful Test Case 19
Table 15: Sign Up Successful Test Case 19
Table 16: Sign Up Unsuccessful Test Case 20
Table 17: Log Out Successful Test Case 21
Table 18: Forgot Password Successful Test Case 22
Table 19: Forgot Password Unsuccessful Test Case 22
Table 20: Resources 27
Table 21: Schedule 27
Table 22: Risks and Assumptions Table 31

1. Introduction

Nekhlawi is an Android application designed to bridge the gap between traditional palm cultivation and care and technology in Saudi Arabia. This app testing plan involves testing and evaluating app features:

1. Account & Profile Management.
2. AI-Based Diagnosis & Treatment Recommendation.
3. Consultation Booking.
4. Consultation Management & Scheduling.
5. Diagnosis & Consultation History Management.
6. Palm Care Knowledge Hub.

1.1 Objectives

The main goal of this test plan is to track the testing processes within the planned schedule to ensure that the application works easily, with good quality and efficiency. All existing features will be tested by using a multi-level testing strategy (Unit, Integration, System) for waterfall SDLC model. The work activity includes Test Case Specification preparation, Test Environment Setup, Execution, Defect Reporting, and the composing of a final Test Summary Report. Budget Resources Needed for Nekhlawi will be determined by Project leader "Zainab Al-Badi". The major milestones will be associated with completion of features and the successful passing of entry/exit criteria for each level of testing. Total testing effort will consume around 30% of the total project development time which is reserved for validation.

1.2 Testing Strategy

The testing strategy of Nekhlawi will ensure the features of software item work well. A V-model approach will be used to ensure comprehensive software quality insurance through the system life cycle.

1. **Test levels:**
   - Unit testing: This phase will help to verify each individual components, methods, classes for Account & Profile Management.
   - Integration testing: This phase expose faults in integrated components. At will

use an incremental approach top-down methodology.

- System testing: This phase will evaluate the end-to-end system and its compliance with all specified functional and non-functional requirements.

2. **Test Techniques and Approach:**
   - White-Box testing: It's a type of Unit testing and Integration test to verify the internal structure and logic of the code. By using the line/statement to ensure every line of code will be executed at least once coverage and basis path test to design test cases to execute independent paths through the program's control flow graph, cyclomatic complexity test.
   - Black-Box testing: To verify functionality based on user requirements without looking at the internal code structure, that will be used during System testing, and UAT.
   - Automated testing: Will be integrated across the development lifecycle, focusing on efficiency and repeatable checks. By using Performance Testing and Regression Testing.
   - Testing Environments: Development Environment: Unit Integration and system testing will be performed within the development environment, using testing frameworks and mock data to isolate components for testing.

3. **Risks and Constrains:**
   - Technical Risks: Due to limited knowledge of integration and connecting AI model, project data, cloud-based services would increase the system failure
   - Data & Accuracy Risks: The chance of the AI model providing a false diagnosis because of the limited images or environmental noise.
   - Timeline Constrain: The project must be completed within specified timeframes.

1.3 Scope

The testing plan will examine all features that exist in the Nekhlawi Android application:

- The testing process will evaluate AI-based systems through their ability to identify pests and diseases in image inputs.
- The testing process will evaluate three functions of the consultation platform which

include real-time chat and video calls and appointment scheduling with agricultural experts.

- The system will verify educational resource accessibility through its ability to present articles and infographics to users.
- The system will verify user account security through its ability to protect account creation and profile updates and history management processes.

The testing plan will not extend beyond its current boundaries which already restrict testing to palm crops and large-scale farm ERP systems.

1.4 Reference Material

- Software Test Plan (STP).
- Software Project Management Plan (SPMP).
- Software Requirements Specification (SRS).
- Software Design Specification (SDS).

1.5 Definitions and Acronyms

Table 1: Definitions and Acronyms

Acronyms	Abbreviation	Definition
STP	Software Test Plan.	It's a formal document used to describe the scope, approaches, activities for software testing.
SPMP	Software Project Management Plan.	It's a formal document used from the beginning and end for managing, controlling the software development.
SRS	Software Requirements Specification.	It's a comprehensive document to outline the software purpose, functions, and constraints of software.

Acronyms	Abbreviation	Definition
STR	Software Test Report.	Its formal report that summarizes the results of the entire testing efforts that have been done in a software project. It provides a detailed account on the coverage of the test, reduction of the success and failure of each test, and the overall quality of the software at a specified time.
SDS	Software Design Specification.	It's a comprehensive document that act as blueprint for system, where it translates the requirements into technical roadmap.
SDLC	Software Development Life Cycle.	A chart illustrates a systems development life cycle that is a summary of the key stages to be undertaken alongside the delivery products in the computerized system validation.
UAT	User Acceptance Testing.	An assessment process whereby the system conformance to established acceptance requirements is followed, allowing the customer to evaluate the compliance of the system to their acceptance levels.
AI	Artificial intelligence.	The computing technology used in the application that will identify pests and diseases by using images as inputs and produce the correct treatment as guidelines.

Acronyms	Abbreviation	Definition
API	Application Programming Interface.	This is a collection of back-end services and methods of exchange between the mobile application, and the cloud or database infrastructures.
CR	Change Request.	An officially written request that is given when an unexpected change in requirements or technical requirements is discovered.
RTM	Requirements Traceability Matrix.	This is a methodical device applied by the development team to assess the effects of the change and to confirm that all the requirements have been tracked to its respective test cases.
ERP	Enterprise Recourse Planning.	Wholesome business management applications: the project specification at Nekhlawi clearly states that the scope of testing will not include large-scale farm enterprise resource planning systems.
VS Code	Visual Studio Code.	Its advanced text editor complete development platform which is a cross-platform Integrated Development Environment (IDE). It serves as the main workspace where developers can write, debug, as well as perform version control of their code.

Acronyms	Abbreviation	Definition
Colab	Google Collaboratory.	It is a Jupyter Notebook on cloud-based environment. It supports literate programming through allowing executable code, rich textual annotations and data visualizations to co-exist together in a single document.
-	Flutter.	Google created the UI (User Interface) Software Development Kit which uses a single codebase to draw native-appearing and acting widgets on both iOS, Android, Web, and desktop platforms.
-	Dart.	It is enacted in a client-streamlined object-oriented programming language and was created to be easy to study besides offering high-performance when it comes to the undertakings that are bound to user interface dialogs.
-	Firebase.	A Business-as-a-Service model of Google. It has a set of cloud-based services including NoSQL databases and authentication systems, which means that developers do not have to maintain their own servers.
-	Super-base.	A BaaS alternative based on open source backends. Despite having similar features to Firebase, it is developed on a PostgreSQL, which is a traditional relational database system.

2. Test Items

This section in the test plan will cover all functional and non-functional components of Nekhlawi system and ensure that all modules comply with the project's Requirements Specification, Design Specification and User Guide.

2.1 Program Modules

In program modules, Nekhlawi Application includes 5 main modules:

1. **User Account & Authentication:** In this module we will test
2. **AI Diagnosis Module:** In this module we will test
3. **Consultation Module:** In this module we will test
4. **Knowledge Hub (Mini Wiki) Module:** In this module we will test
5. **History Module:** In this module we will test

2.2 Job Control Procedures

Not applicable to the Nekhlawi system.

2.3 User Procedures

Testing the User Guide includes:

1. Reviewing it to ensure it clear explanation of all system functions.
2. Ensures the including of all the screenshots that match the actual interface.
3. Verifying instructions are accurate, up-to-date, and easy to follow.
4. Checking that all steps are complete and no essential guidance is missing.

2.4 Operator Procedures

Not applicable to the Nekhlawi system.

3. Features to Be Tested

Table 2: Features To Be Tested

Req ID	Feature	Applicable Role	Priority (1-5)	Description
User Account & Authentication Module				
1	Role Selection	Both (Expert user – Non expert user)	5	The user must choose the type of account (an expert user who provides consultations - an unexpert user who reserves the consultations) before creating an account and based on his selection, the fields are displayed to be filled out by the user
2	Sign-up	Both	5	After choosing a role, the user will fill in the fields and after that a new account will be created and saved in the data base
3	Log-in	Both	5	The user logs in to the application using the registered email and the password, a magic link is sent to the registered email, when the user clicks on it, the system redirects the user to the home page in the application and displays its data
4	Log-out	Both	5	The user can log out successfully from his accounts, the system redirects the user to the sign up page

Req ID	Feature	Applicable Role	Priority (1-5)	Description
User Account & Authentication Module				
5	Forget Password	Both	5	The user can change the password before logging in. The user enters the registered email, and the system will send a magic link to the email, when the user clicks on it, the system redirects the user to the page of changing the password inside the application and chang the password
6	Delete Profile	Both	5	The user can delete his account and all his data from the data base completely
7	Profile Management	Both	5	The user can update his profile information and change the account password
8	Error Messages Clarity	Both	4	If the user enters any data that system shall validate user input and display an appropriate error message if the entered data does not match the required format
9	Image Upload / Camera Capture	Both	4	The user can upload or capture clear palm images for diagnosis successfully
10	AI Disease Detection	Both	5	AI model analyzes the uploaded image and returns the predicted disease

Req ID	Feature	Applicable Role	Priority (1-5)	Description
AI Diagnosis Module				
11	Result Display Page	Both	4	The system displays the AI prediction to the user
Consultation Module				
12	Booking Consultation	Non expert user	4	Non experts can schedule consultation sessions
13	Chat System	Both	4	Experts and non experts in the start of the scheduled can communicate with experts via real-time chat
14	Notification System	Both	2	The user receives reminder notifications before the consultation sessions begin and for messages and booking updates
15	Consultation Management	Both	3	The user can modify the consultation appointment
16	View Expert Profile	Non expert user	3	The user can view all the available experts and view their profiles (expert name, specialization, years of experience, bio, rating and past review)
17	Submit Rating	Non expert user	3	After the end of the consultation session user rate the expert and the session and can optionally write a review

Req ID	Feature	Applicable Role	Priority (1-5)	Description
Knowledge Hub (Mini Wiki) Module				
18	Article Display	Both	4	The user can view all the articles in the DB
19	Search Articles	Both	1	The user can search in the articles by the article title
History Module				
19	View Consultation History	Both	3	The user can view his past consultations
20	View Diagnosis History	Both	3	The user can view his past AI diagnosis
21	Search Consultation History	Both	1	The user can search in his history for any past consultations
22	Search Diagnosis History	Both	1	The user can search in his history for any past AI diagnosis

4. Features Not to Be Tested

Table 3 : Features Not to Be Tested

Feature	Justification
Unsupported Devices & Versions	Nekhlawi application does not support outdated Android versions and older devices (Android 12 and older); hence, the testing plan will not cover them.
Third-Party Platform Failure	Our application will only integrate a mock of the external services. The failures actually caused by external third-party services, such as cloud services, will not be covered by our testing plan.
Cyber Attacks Simulation	The Nekhlawi app development team consists of students with limited resources and time constraints to implement advanced security testing.
Extreme Performance & Stress Testing	Stress testing with a high load of simultaneous users is not needed, as the target users are expected to be limited.

5. Approach

The testing process of Nekhlawi Mobile Application will be through systematic and scheduled activities to verify both functional and non-functional software requirements which should enable the proper estimation for completing each activity using both black box and white box methodologies.

Main testing activities will include: Component Testing, Integration Testing, Interface Testing, Security Testing, Recovery Testing, Performance Testing, Regression Testing and Acceptance testing.

where each is defined with the objectives, inputs, methods, criteria, and expected outputs. The tests will be done manually and with the help of some testing tools. This will be achieved through the use of a Requirement Traceability Matrix (RTM) to ensure requirement traceability, where each requirement is accompanied by what test cases and test results. This type of testing is used to make sure that all the activities of the system are addressed, defects are found and detected with minimal effort and that Nekhlawi meets performance, security, and user acceptability requirements during the release.

5.1 Component Testing

Component testing (unit testing) aim is to verify that each distinct module and unit functions correctly, while also confirming the accuracy, completeness, and alignment of the units with the design specifications. The development team will carry out the component testing by building test cases for white box testing methods on each individual module source code. A method of basis path testing will be applied to guarantee that every independent branch at each decision point is executed by using tools such as JUnit.

- **AI Diagnosis Module**

Table 4: Ai Diagnosis Module Successful Test Case

Successful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
A 5MB clear palm leaf Image from user's gallery	AI diagnosis & confidence rate	-	-	The diagnosis will be saved in history

Table 5: Ai Diagnosis Module Unsuccessful Test Case

Unsuccessful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
A captured palm leaf image in darkness	Message: "Image unclear. Unable to determine diseases accurately." And offers retaking the image or booking consultation	-	-	The diagnosis won't be saved in history

Unsuccessful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
A 23MB palm leaf image from user's gallery	Message: "Image size is too large. please upload an image under 10MB"	-	-	

- **Consultation Module: Booking**

Table 6: Consultation Module: Booking Successful Test Case

Successful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
User clicks at expert > Expert available > Slot available > Connected > Payment Success	Mark booking as pending, waiting expert approval	-	-	-

Table 7: Consultation Module: Booking Unsuccessful Test Case

Unsuccessful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Select an unavailable expert	Message: "No expert available at the moment"	-	-	-
Select an unavailable time slot	"Selected time is no longer available. Please pick another slot"	-	-	-

Unsuccessful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Lost Connection while clicking “Confirm booking”	Message:“Unable to complete booking. Check your connection and try again”	-	-	-
Paying with insufficient funds	Message: “Unable to complete booking. Due to the payment is not complete”	-	-	-

- **Consultation Module: Management**

Table 8: Consultation Module: Management Successful Test Case

	Successful Test Case				
	Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Expert	Click “Approve booking” on a pending request	User receives "Consultation successfully approved" alert.	-	-	Status: "Approved"
	Click “Decline” on a pending request	User receives reschedule/selection prompt.	-	-	Status: "Declined"
	No consults	Display: "You don't have any upcoming consultations at the moment."	-	-	-
Farmer	View notification > click “Cancel”	Display confirmation prompt	-	-	Status "Canceled"
	View notification > clicks “Cancel” > clicks “Back”	Process stays in "Incoming chat" state	-	-	No status change
	View notification > clicks “Reschedule”	Redirect to "Book Consultation” page	-	-	Status "Rescheduled"

- **Consultation Module: Chat System**

Table 9: Consultation Module: Chat System Successful Test Case

Successful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Wait for session time > type “hello”, > send	Typed message sent and reached the other party	-	-	-
Click “+” to add file > permission to gallery granted > upload “image.jpg”	Image is successfully uploaded and displayed in the chat.	-	-	-
Click "End Session"> click “Confirm” > submit rating	Chat closes and review saved in the data base	-	-	Status: "Complete"

Table 10: Consultation Module: Chat System Unsuccessful Test Case

Unsuccessful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Tap "Chat Consultation" before scheduled time	Chat remains locked/inactive	-	-	-
Click “+” to add file > Deny access to gallery	Message: "Media access was denied. You cannot send photos..."	-	-	-
Click “+” to add file > permission to gallery granted > upload “mypalm.exe”	Message "Unsupported file format. Please select a valid file."	-	-	-

Unsuccessful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Click “End session” > click “Cancel”	Conformation prompt closes and the chat session stays active	-	-	-

- **Knowledge Hub (Mini Wiki) Module**

Table 11: Knowledge Hub (Mini Wiki) Module Successful Test Case

Successful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Click at Article #1	Article #1 Displayed	-	-	-

- **History Module**

Table 12: History Module Successful Test Case

Successful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
User selects “history” icon > picks past Ai diagnoses> clicks “Export”	Display past diagnoses details (image/result/date) and exports the file	-	-	-
User selects “history” icon > picks past consultations > clicks “Export”	Display past consultations details (expert notes/date/ status) and exports the file	-	-	-

Table 13: History Module Unsuccessful Test Case

Unsuccessful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
User selects “history” icon with no history	Display empty page with text “No diagnosis or consultations found yet, start with scanning your palm tree or book consultation”	-	-	-
User selects “history” icon > picks past Ai diagnoses or consultation	Displays past Ai diagnosis/consultations details	-	-	“only viewing”
Click item > click “Export”>click “Cancel” on conformation	Export process is terminated and redirect to the selected history page.	-	-	-

5.2 Integration Testing

The primary objective of Integration Testing is to ensure that distinct software modules within Nekhlawi system which have previously undergone testing and operate correctly when interfaced with other subsystems. It is important to ensure that the interfaces of the modules are performing as intended and that the overall system functions in accordance with the design and operational specifications. The methodology adopted will be Black box testing using the incremental approach or the top-down methodology and will be applied using tools such as Postman and Firebase Console.

- **Role selection**

Table 14: Role selection Successful Test Case

Successful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Clicks "Expert/Farmer" role button	Displays Expert/Farmer account creation form according to role	-	-	-

- **Sign up**

Table 15: Sign Up Successful Test Case

	Successful Test Case				
	Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Expert	Name: Ahmed Sumpul Email:AhmedSumpul@gmai.com Phone Number:9667329123 Years of Experince: 10 Specialty: Pests Password: 99223338S$ Confirm password: 99223338S$ Terms and conditions: accepted Captcha: accepted Code: 9999	Display Dashboard page	-	-	New Account information stored in database

	Successful Test Case				
	Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Farmer	Name: Mohammed Ahmed Email: Moh67@gmail.com Phone Number: 966 1234567 Password: 12345678B# Confirm password: 12345678B# Terms and conditions: accepted Captcha: accepted Code: 9999	Display Dashboard	-	-	-

Table 16: Sign Up Unsuccessful Test Case

Unsuccessful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Name: Mohammed Ahmed Email: **Moh67@gm//ail.com** Phone Number: 966 1234567 Password: 12345678B# Confirm password: 12345678B# Terms and conditions: accepted Captcha: accepted Code: 9999	Redirect to error field with error indication	-	-	-

Unsuccessful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Name: Mohammed Ahmed Email: Moh67@gmail.com Phone Number: 966 1234567 Password: 12345678B# Confirm password: 12345678B# Terms and conditions: accepted Captcha: accepted **Code: Invalid**	Message: "Invalid or expired code"			
Email: Moh67@gmail.com Phone Number: 966 1234567 Password: 12345678B# Confirm password: 12345678B# Terms and conditions: accepted **Captcha: Unaccepted** Code: Invalid	Message: "Failed Captcha, redo again"			

- **Log Out**

Table 17: Log Out Successful Test Case

Successful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Clicks "log out" button	The user will exit the account and redirected to "log in" screen	-	-	-

- **Forgot Password**

Table 18: Forgot Password Successful Test Case

Successful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Email: AhmedSumpul@gmail.com Completed Capcha Code: 0233 New password: 99223338S$	Message: "Password successfully reset"	-	-	-

Table 19: Forgot Password Unsuccessful Test Case

Unsuccessful Test Case				
Input	**Expected Output**	**Actual Output**	**Result**	**Comment**
Email: Ahmed??a@gmail.com	Message: "Email not found"	-	-	-
Email: AhmedSumpul@gmail.com Uncompleted Captcha Code: 0233 New password: 99223338S$	Message: "Failed Captcha, redo again"	-	-	-
Email: AhmedSumpul@gmail.com Completed Captcha Wrong Code	Message: "Invalid or expired code"	-	-	-
Email: AhmedSumpul@gmail.com Completed Captcha Code: 0233 New password: u99	Display error notification and password rules	-	-	-

5.3 Conversion Testing

Not Applicable.

5.4 Job Stream Testing

Not Applicable.

5.5 Interface Testing

Interface testing will be conducted to ensure that the Nekhlawi backend interacts effectively and as intended with the external systems such as payment gateways and account authentication services. It will be executed by the testing and the development team after the integration process with the external systems where it will involve the execution of API calls, the simulation of responses and the verification of error handling through applying tools such as Espresso.

5.6 Security Testing

Security testing is performed to ensure that the Nekhlawi system is able to enforce proper access control and validate being free of any major security vulnerabilities. The test will be performed by the quality assurance and testing teams after integration by using password vulnerability tests and role checks using tools like Firebase Console and Postman.

5.7 Recovery Testing

Recovery test will be conducted to ensure that Nekhlawi system recovery and backup processes work as intended to when any failures occur. The test will be conducted by the Testing Team after the backup procedures are configured by simulating failures (server crash, network loss) and ensuring the proper system recovery by using tools like Firebase Console.

5.8 Performance Testing

Performance testing is a process that evaluates the system efficiency of the Nekhlawi system while operating under different levels of user interaction. This is aimed at measuring the response time

of the system, the stability of the system, and whether it can handle simultaneous interactions among users without delays and malfunctions. This will determine whether the system is performing within the set performance standards and the test scenarios will be used to measure the time needed by the system to respond to various activities, including navigations, booking, and checkouts.

5.9 Regression Testing

Regression testing is conducted to verify that any changes or enhancements made to the Nekhlawi system will not adversely impact features that have been already tested / were validated to function correctly to preserve the system's stability after the introducing any new functionalities, in appearance of any bugs, or modifications to any segments of the code or database. The test will be conducted using tools such as GitHub, Junit and Espresso.

5.10 Acceptance Testing

Acceptance testing is conducted as a component of quality assurance to verify that Nekhlawi satisfies all of its specified requirements, ensures that it operates as intended from the perspective of the end-user, and identifies defects and issues while it’s still possible. Prior to deployment, the Product Owner will implement specified acceptance test cases aligned with SRS requirements, where End users engage in realistic scenarios (such as entering mini-wiki, reviewing history, utilizing AI diagnosis, and scheduling a consultation). These tests will transpire within a controlled User Acceptance Testing (UAT) environment that emulates the production setting and will culminate in the confirmation from end users. All acceptance criteria established within the SRS must be fulfilled without any outstanding critical defects.

5.11 Beta Testing

Not Applicable.

6. Pass / Fail Criteria

The specified criteria for accepting or rejecting test case results are listed below:

- Pass: The result of test case execution matches the expected result, and the function works

as specified in the system's requirements. Data is stored properly in the database and protected from unauthorized access.

- Fail: The test case execution result does not match the expected output; data is stored incorrectly, the function does not perform properly, or any other system deviation from the intended output occurs.

6.1 Suspension Criteria

Testing shall be suspended under the following conditions:

- Missing Components: Major parts of the system have not been built yet, leading to inaccurate testing results or making testing infeasible.
- Critical Defects: Defects that prevent the execution of the test process sequence, such as a login failure.
- Backend Services Inaccessibility: Database connection is lost, or core API services do not respond.
- AI Model Accuracy: The confidence score of the AI model does not meet the required threshold point.

6.2 Resumption Criteria

Testing shall resume when the following conditions are met:

- sBuilt Components: Functions and code blocks necessary for testing have been built.
- Defect Resolution: The Quality Assurance team has verified that blocking defects are resolved.
- Backend Services Accessibility: All needed services are responding to the system's requests.
- AI Model Accuracy: The confidence score of the AI model meets the threshold point.

6.3 Approval Criteria

The following conditions must be met to approve the test results:

- Requirement Fulfillment: All system features mentioned in the SRS work as intended.

- Success Rate: At least 90% of the critical test cases succeed.
- Defect Density: Less than 10% of non-major defects are detected.
- Performance: The system meets benchmarking performance standards.
- Formal Sign-off: Formal approval from the stakeholder is received.

7. Testing Process

This section discusses the expected documentation, tasks to be completed, resources, responsibilities, and schedule to be followed.

7.1 Test Deliverables

By the end of the testing activities, the two documents below should be delivered:

- Software Test Plan (STP).
- Software Test Report (STR).

7.2 Testing Tasks

- Develop all required documents during the testing process
- Prepare the testing environment and assign resources and tools
- Define testing tasks, responsibilities and schedules
- Educate on testing cases and activities using the proper approaches
- Document and resolve defects
- Assure the quality of the software and check for compliance with requirements
- Manage any changes that occur

7.3 Responsibilities

Using the guidance of the team leader, all team members will be responsible for planning and conducting testing activities, coordinating resources, and following the schedule. Moreover, team members will be responsible for developing and assuring the quality of the software by resolving defects and verifying code quality. On the other hand, the academic supervisor is responsible for supervising and approving overall processes.

7.4 Resources

Table 20: Resources

Requirements	Resources
Human	• Team members • Supervisor
Hardware	• Personal laptops • Cloud-based processing
Software	• VS Code • Google collab • APIs • Libraries • Android studio • Firebase • Super-base • Google Drive • Flutter • Dart

7.5 Schedule

Table 21: Schedule

Testing Task	Start	End
Plan Development	08/02/2026	18/02/2026
Test Case Preparation	12/04/2026	16/04/2026
Setting Test Environment Components	19/04/2026	21/04/2026
Integration Testing	22/04/2026	24/04/2026

Testing Task	Start	End
System Testing	25/04/2026	30/04/2026
Regression Testing	01/05/2026	03/05/2026
Performance Testing	04/05/2026	06/05/2026
Security Testing	07/05/2026	08/05/2026
Recovery Testing	08/05/2026	09/05/2026
Acceptance Testing	10/05/2026	12/05/2026
Report Completion	13/05/2026	16/05/2026

8. Environmental Requirements

8.1 Hardware

The following hardware resources are required:

- Android smartphones (13 and above versions)
- high-performance computers
- Stable internet access
- Backup storage device for Nakhlawi application data
- Cloud server infrastructure

8.2 Software

The following software resources are required:

- Android Studio
- Firebase / Cloud backend services
- AI model framework (Tensorflow)
- Version control system (GitHub)
- Testing frameworks (JUnit, Espresso)
- Browser (Google Chrome / Edge) for backend dashboard testing.
- Database management system (Firebase Firestore or MySQL)

8.3 Security

The following security measures are required during testing:

- Backend systems access control
- Secure database transmission (HTTPS/SSL)
- Restricted access to AI model configuration files
- Test database backup and recovery mechanisms
- Secrecy of test user data to avoid unauthorized modification

8.4 Tools

The following tools will be used:

- **JUnit**

  It is used to perform unit testing of the Java components and check the logic of the

program to verify that the various modules are working properly.

- **Espresso**

  It is applied to test the interfaces of Android applications automatically to test the interactions between users.

- **Postman**

  Applied in API testing and verification of the backend services to ensure that the requests and responses are handled appropriately.

- **Firebase Console**

  To trace database activities, authentication, and backend log testing.

- **GitHub**

  Used to track changes, version control, and collaborative development management.

- **Performance Monitoring Tools**

  It is used to test response time, load performance of a system, and general application performance.

- **Microsoft Word**

  Test plans, test case specifications, defect reports, and final test summary reports are prepared and documented using it.

Training and Support:

The setup and efficient use of all testing tools will be assisted through official documentation, online tutorials, and user manuals by team members.

8.5 Publications

- Software Design Specification (SDS)

- Software Requirements Specification (SRS)
- API Documentation
- AI Model Documentation
- Android Development Documentation
- User Guide / User Manual

8.6 Risks and Assumptions

Table 22: Risks and Assumptions Table

#	Risks	Assumptions	Contingency Plans
1	**Low AI Accuracy** The AI model can be used to give wrong or low-confidence diagnoses because of limited training data or low-quality images	System testing is preceded by training the AI model on adequate and appropriate palm disease datasets	Expand the testing data size and range, change the level of confidence, and retrain or optimize the model (where needed)
2	**Time Constraints** The defect rework or integration delay can cause the testing activities to exceed the planned schedule	The development modules will all be accomplished on schedule	Select the important and risky test cases, use automated regression testing, and change the testing schedule where necessary
3	**Limited Testing**	The testing phase will be able to access required hardware	Test Android Emulator with various settings and

	Devices Lack of Android devices can limit compatibility tests with various settings	resources	get other devices where possible
4	**Backend Service Downtime** APIs or cloud-based services can go unavailable at certain points of testing	Backend infrastructure and APIs will not be changed during testing and will be stable and accessible	Keep reserve test data and set up offline testing procedures that do not need to be connected to the cloud

9. Change Management Procedures

9.1 Change initiation:

When we identify a chance of unplanned modification of Nekhlawi main requirements, or technical, we will prepare the change request document (CR) and then send it to Mrs. Albandari for approval throw the progress report in week seven.

9.2 Change review:

Once the CR is complete, the team will analyze the effect of the change by using the Requirements Traceability Matrix (RTM), identify any resources needed, and be on track with the planned schedule.

9.3 Change authorization process:

The project team will begin the Technical Realignment phase after we receive formal

approval of the Change Request (CR). The development branch will undergo changes through the addition of new features and the removal of existing ones. The team of Quality Assurance will conduct Test Suite Synchronization while developing new test cases for new features and deactivate test cases which test obsolete functions.

10. Plan Approvals

This plan is approved by:

- Project Leader: Zainab Hassan Al-Badi
  Date: 1 / 3 / 2026
  Signature:

- Project Supervisor: Mrs. Albandary Mohammad Alamer
  Date: 1 / 3 / 2026
  Signature: